

CSE110 | Programming Language - I

Quiz: 1

Name:	ID:	Sec:
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The HR department of Ami110PariNa Ltd. wants to automate their yearly employee evaluation process. Every year, they assess employees based on performance score, years of service, and attendance to decide whether an employee is eligible for a bonus and/or promotion. The HR department follows these rules:

1. If the performance score is 80 or above, the employee qualifies for a bonus.
 - If the employee has worked for 5 or more years, they get a 10% bonus.
 - Otherwise, they get a 5% bonus.
2. If the performance score is below 80, the employee does not get any bonus.
3. Regardless of the bonus:
 - If attendance percentage $\geq 95\%$ and performance score ≥ 85 , the employee also gets a promotion.
 - Otherwise, no promotion.

Draw the flowchart for the above scenario. Make sure to identify the inputs, outputs properly.

[NOTE: BASE SALARY FOR EVERYONE IS 25,000 Taka]

BRAC UNIVERSITY (SET-A)

CSE110 Exam: Quiz 02

Semester: Spring26

Full Marks: 10

Duration: 20 Mins

Name:	ID:	Section:
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Question 1 [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (* or /) from the user. Your task is to create two numbers from the two input strings by considering only the odd digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than .length() and .charAt() methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q *	371 531 197001
5dQ!87 26f0&3 /	57 3 19
Explanation: For sample Input 1, ab4s3F67h%21 contains odd digits 3,7,1, that's why the first number is 371 mnP543&1b86Q contains odd digits 5,3,1, that's why the second number is 531 After performing the addition operation (*), the final output is 197001	

Solution of Set A:

import java.util.Scanner;		
public class quiz{	class	[0.5 marks]
public static void main(String [] args){		
Scanner sc= new Scanner(System.in);	2 input	[0.5 marks]
String string1=sc.next();		
String string2=sc.next();		
String num1="";		
for(int i=0;i<string1.length();i++){	odd digit concat->String 1	[2 marks]
int ascii = string1.charAt(i);		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num1+=string1.charAt(i);		
}		
}		
String num2="";		
for(int i=0;i<string2.length();i++){	odd digit concat->String 2	[2 marks]
int ascii = string2.charAt(i); //i=0 ascii=97		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num2+=string2.charAt(i);		
}		
}		
int power= num1.length()-1;	string 1->number 1	[2 marks]
int new_num=0;		
for(int i=0;i<num1.length();i++){		
int ascii=num1.charAt(i);		
int num=ascii-48;		
new_num+=num* Math.pow(10,power);		
power--;		
}		
int power2= num2.length()-1;	string 2->number 2	[2 marks]
int new_num2=0;		
for(int i=0;i<num2.length();i++){		
int ascii=num2.charAt(i);		
int num=ascii-48;//num=1		
new_num2+=num* Math.pow(10,power2);		
power2--;		
}		
String operator=sc.next();	* or /	[1 mark]
if(operator.equals("")){		
System.out.println(new_num*new_num2);		
}		
else{		
System.out.println(new_num/new_num2);		
}		
}}		

BRAC UNIVERSITY (SET-B)

CSE110 Exam: Quiz 02

Semester: Spring26

Full Marks: 10

Duration: 20 Mins

Name:	ID:	Section:
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Question 1 [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (+ or -) from the user. Your task is to create two numbers from the two input strings by considering only the odd digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than `.length()` and `.charAt()` methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q +	371 531 902
5dQ!87 26f0&3 -	57 3 54
Explanation: For sample Input 1, ab4s3F67h%21 contains odd digits 3,7,1, that's why the first number is 371 mnP543&1b86Q contains odd digits 5,3,1, that's why the second number is 531 After performing the addition operation (+), the final output is 902	

Solution of Set B:

import java.util.Scanner;		
public class quiz{	class	[0.5 marks]
public static void main(String [] args){		
Scanner sc= new Scanner(System.in);	2 input	[0.5 marks]
String string1=sc.next();		
String string2=sc.next();		
String num1="";		
for(int i=0;i<string1.length();i++){	odd digit concat->String 1	[2 marks]
int ascii = string1.charAt(i);		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num1+=string1.charAt(i);		
}		
}		
String num2="";		
for(int i=0;i<string2.length();i++){	odd digit concat->String 2	[2 marks]
int ascii = string2.charAt(i); //i=0 ascii=97		
if((ascii>48&&ascii<=57)&&(ascii%2!=0)){		
num2+=string2.charAt(i);		
}		
}		
int power= num1.length()-1;	string 1->number 1	[2 marks]
int new_num=0;		
for(int i=0;i<num1.length();i++){		
int ascii=num1.charAt(i);		
int num=ascii-48;		
new_num+=num* Math.pow(10,power);		
power--;		
}		
int power2= num2.length()-1;	string 2->number 2	[2 marks]
int new_num2=0;		
for(int i=0;i<num2.length();i++){		
int ascii=num2.charAt(i);		
int num=ascii-48;//num=1		
new_num2+=num* Math.pow(10,power2);		
power2--;		
}		
String operator=sc.next();	* or /	[1 mark]
if(operator.equals("+")){		
System.out.println(new_num+new_num2);		
}		
else{		
System.out.println(new_num-new_num2);		
}		
}}		

Set-A

Write a Java program that does the following:

1. Take two strings as input from the user (e.g., string1 and string2), which may contain a mix of letters, digits, and special characters (e.g., "ab4s3F67h%21").
2. From each string, extract only the **odd digits** (digits that are odd numbers, i.e., 1, 3, 5, 7, 9) — ignore all letters, special characters, and even digits.
3. Combine the extracted odd digits (in the order they appear) to form a new number for each string (e.g., if the odd digits found are '3', '7', '1', the number formed should be 371).
4. Print both newly formed numbers, separated by a comma.
5. Take a third input representing a **mathematical operator** (either * or /).
6. Based on the operator entered:
 - If it's *, print the **product** of the two numbers.
 - Otherwise (assume /), print the **quotient** (integer division) of the two numbers.

Set-B

Write a Java program that performs Selection Sort on an integer array.

Given an array {3, 7, 2, 29, 38, 59, 1}:

1. Implement **Selection Sort** to sort the array in ascending order:
 - For each position i in the array, find the index of the **minimum element** in the unsorted portion of the array (from i to the end).
 - Swap that minimum element with the element at position i.
2. Print the fully sorted array using Arrays.toString().

BRAC UNIVERSITY (SET-A)

CSE110 Exam: Quiz 03

Semester: Spring26

Full Marks: 20

Duration: 20 Mins

Name:	ID:	Section:
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Question 1 [20 Points]

A weekly attendance system stores 7 digits in an array, where **each index** represents a day of the week in order from Saturday to Friday. In this system, classes on **Saturday** (index 0) and **Tuesday** (index 3) are conducted **online**, while all other days are conducted **in-person**. A day is considered **balanced** if the number of elements before it (to the left) with lower values is equal to the number of elements after it (to the right) with higher values.

Write a Java program to determine and print **all** balanced days **only among the in-person days**, and also print the **total** number of such balanced in-person days.

Given Array	Output
<pre>int[] arr = {40, 10, 20, 100, 15, 30, 5};</pre>	Balanced in-person days are: 2 4 6 Total balanced in-person days: 3
Explanation: Monday (index 2, value 20): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Wednesday (index 4, value 15): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Friday (index 6, value 5): left lower count = 0, right higher count = 0, ignoring indices 0 and 3 (online). Since both are equal, the day is balanced.	

Solution:

public class quiz03{		
public static void main(String [] args){		
int [] arr = {40, 10, 20, 100, 15, 30, 5};	2 arrays initialisation	[2 marks]
int [] arr2 = new int[arr.length];		
int count=0;	3 variables	[1.5 marks]
for(int i=0;i<arr.length;i++){	loop	[0.5 marks]
int left=0,right=0;		
if(i==0 i==3){	online day skip	[2 marks]
continue;		
}		
for(int j=i+1;j<arr.length;j++){	right count	[4 marks]
if(!(j==0 j==3)){		
if(arr[j]>arr[i]){		
right++;		
}		
}		
}		
for(int j=i-1;j>0;j--){	left count	[4 marks]
if(!(j==0 j==3)){		
if(arr[j]<arr[i]){		
left++;		
}		
}		
}		
if(left==right){	count increase, new array update	[3 marks]
arr2[count]=i;		
count++;		
}		
}		
System.out.print("Balanced in-person days are: ");	print	[0.5 mark]
for(int i=0;i<count;i++){	print new array	[2 marks]
System.out.print(arr2[i]+" ");		
}		
System.out.println("\nTotal balanced in-person days: "+count);	print	[0.5 mark]
}		
}		

BRAC UNIVERSITY (SET-A)

CSE110 Exam: Quiz 03

Semester: Spring26

Full Marks: 20

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Name:	ID:	Section:
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Question 1 [20 Points]

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Write a Java program to determine and print **all** balanced days **only among the in-person days**, and also print the **total** number of such balanced in-person days.

Given Array	Output
<pre>int[] arr = {40, 10, 20, 100, 15, 30, 5};</pre>	Balanced in-person days are: 2 4 6 Total balanced in-person days: 3
Explanation: Monday (index 2, value 20): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Wednesday (index 4, value 15): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Friday (index 6, value 5): left lower count = 0, right higher count = 0, ignoring indices 0 and 3 (online). Since both are equal, the day is balanced.	

Solution:

public class quiz03{		
public static void main(String [] args){		
int [] arr = {40, 10, 20, 100, 15, 30, 5};	2 arrays initialisation	[2 marks]
int [] arr2 = new int[arr.length];		
int count=0;	3 variables	[1.5 marks]
for(int i=0;i<arr.length;i++){	loop	[0.5 marks]
int left=0,right=0;		
if(i==0 i==3){	online day skip	[2 marks]
continue;		
}		
for(int j=i+1;j<arr.length;j++){	right count	[4 marks]
if(!(j==0 j==3)){		
if(arr[j]>arr[i]){		
right++;		
}		
}		
}		
for(int j=i-1;j>0;j--){	left count	[4 marks]
if(!(j==0 j==3)){		
if(arr[j]<arr[i]){		
left++;		
}		
}		
}		
if(left==right){	count increase, new array update	[3 marks]
arr2[count]=i;		
count++;		
}		
}		
System.out.print("Balanced in-person days are: ");	print	[0.5 mark]
for(int i=0;i<count;i++){	print new array	[2 marks]
System.out.print(arr2[i]+" ");		
}		
System.out.println("\nTotal balanced in-person days: "+count);	print	[0.5 mark]
}		
}		

Question 1 [20 Points]

Write a method called `rectPerimeter` that takes 4 integer sides as arguments.

- If the sides can form a valid rectangle, calculate and print the perimeter.
- Otherwise, print "Not a Rectangle".

Sample Input	Sample Output
<code>rectPerimeter(5,7,5,7);</code>	24
<code>rectPerimeter(3,4,5,6);</code>	Not a Rectangle

```
public class quiz04{  
    2 marks  
    public static void main(String [] args){  
        2 marks  
        rectPerimeter(5,7,5,7);  
        2 marks  
        rectPerimeter(3,4,5,6);  
        2 marks  
    }  
  
    public static void rectPerimeter(int a,  
    int b, int c, int d){    2 marks  
        if(a==c && b==d){  
            5 marks  
            System.out.println(a+b+c+d);  
        }  
        Else{  
            5 marks  
            System.out.println("Not a  
Rectangle");  
        }  
    }  
}
```